JEE APEX FTS – JEE ADV. FULL SYLLABUS TEST-04 [P2]

KEY SHEET

MATHEMATICS

1	**C**	2	**A**	3	**A**	4	**C**	5	**ABCD**	6	**BC**
7	**BC**	8	**9**	9	**22**	10	**0**	11	**5**	12	**2**
13	**8**	14	**4.5**	15	**4**	16	**0.6**	17	**16**		

PHYSICS

18	**A**	19	**B**	20	**B**	21	**B**	22	**BC**	23	**BD**
24	**B**	25	**3**	26	**2**	27	**3**	28	**7**	29	**5**
30	**4**	31	**4**	32	**7.2**	33	**6**	34	**20**		

CHEMISTRY

35	**C**	36	**B**	37	**A**	38	**D**	39	**AC**	40	**ABC**
41	**AD**	42	**4**	43	**2**	44	**6**	45	**80**	46	**12**
47	**108**	48	**0.97-0.99**	49	**0.16-0.18**	50	**5**	51	**1**		

SOLUTIONS
MATHEMATICS

1. Volume $= \left|\left[2\vec{b}\times\vec{c}, 3\vec{c}\times\vec{a}, 4\vec{a}\times\vec{b}\right]\right| = 18$

$24\left[\vec{a}\,\vec{b}\,\vec{c}\right]^2 = 18 \Rightarrow \left|\left[\vec{a}\,\vec{b}\,\vec{c}\right]\right| = \frac{\sqrt{3}}{2}$

Now, $\left[\vec{a}\,\vec{b}\,\vec{c}\right] = \begin{vmatrix} (1+\sin\theta) & \cos\theta & \sin 2\theta \\ \sin\left(\theta+\frac{2\pi}{3}\right) & \cos\left(\theta+\frac{2\pi}{3}\right) & \sin\left(2\theta+\frac{4\pi}{3}\right) \\ \sin\left(\theta-\frac{2\pi}{3}\right) & \cos\left(\theta-\frac{2\pi}{3}\right) & \sin\left(2\theta-\frac{4\pi}{3}\right) \end{vmatrix}$

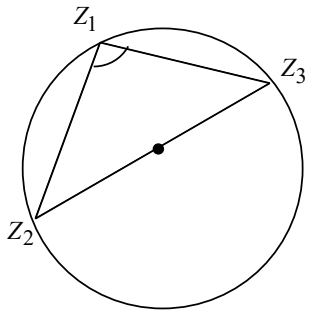
Applying $R_1 \to R_1 + R_2 + R_3$ we get after simplification,

$\left|\left[\vec{a}\,\vec{b}\,\vec{c}\right]\right| = |\cos 3\theta| = 1$

2. $Z_1.\left(\overline{Z}_2 + \overline{Z}_3\right) + \overline{Z}_1\left(\overline{Z}_2 + \overline{Z}_3\right) = 0$

$\Rightarrow |Z_1|^2 + |Z_2|^2 - Z_1.\overline{Z}_2 - \overline{Z}_1.Z_2 + |Z_1|^2 + |Z_3|^2 - Z_1.\overline{Z}_3 = 4$

$\Rightarrow (Z_1 - Z_2).\left(\overline{Z}_1 - \overline{Z}_2\right) + (Z_1 - Z_3).\left(\overline{Z}_1 - \overline{Z}_3\right) = 4 = 2^2$



Hence Z_2, Z_3 are diametrically end points. $\Rightarrow Z_2 + Z_3 = 0 \Rightarrow |Z_1 + Z_2 + Z_3| = 1$

3. $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ n & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 2^{-1} & 2^{-2} & 2^{-n} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\Rightarrow A^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ n & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 2^{-1} & 2^{-2} & 2^{-n} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ n & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 2^{-1} & 2^{-2} & 2^{-n} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$\Rightarrow A^2=\begin{bmatrix}1&0&0&0\\2&0&0&0\\3&0&0&0\\n&0&0&0\end{bmatrix}\begin{bmatrix}\sum i.2^{-i}&0&0&0\\2&0&0&0\\3&0&0&0\\n&0&0&0\end{bmatrix}\begin{bmatrix}2^{-1}&2^{-2}&2^{-n}\\0&0&0\\0&0&0\\0&0&0\end{bmatrix}=\sum i.2^{-i}.A$$

$$\Rightarrow A^2=A.\sum i.2^{-i}\Rightarrow A^n=\left(\sum i-2^{-i}\right)^{n-i}A\ \Rightarrow tr\left(A^n\right)=\left(\sum i.2^{-i}\right)^n$$

$$\Rightarrow\left(tr\,A^n\right)^{1/n}=\sum i.2^{-i}=\frac{1}{2}+\frac{2}{4}+\frac{3}{8}+-----\infty=2$$

4. $\left(\sqrt{3}+1\right)^6=I+F$ where $0\le F<1$

Let $\left(\sqrt{3}-1\right)^6=G$ where $0\le G<1$ $\quad\therefore I+F+G=\left(\sqrt{3}+1\right)^6+\left(\sqrt{3}-1\right)^6$

$=2\left\{{}^6C_0\left(\sqrt{3}\right)^6+{}^6C_2\left(\sqrt{3}\right)^4+{}^6C_4\left(\sqrt{3}\right)^2+{}^6C_6\right\}$ $\quad=2\{1.27+15.9+15.3+1\}=416$

or $I=416-\left(F+G\right)=415$ $\quad$ (AS O < F + G < 2)

5. $\alpha+\beta=\tan\theta-2$

$\alpha.\beta=-\tan\theta-1\quad\Rightarrow\alpha+\beta+\alpha\beta=-3\ \Rightarrow\quad(1+\alpha).(1+\beta)=-2$

since α,β are integers possible values can be

(i) $1+\alpha=2,\quad 1+\beta=-1$

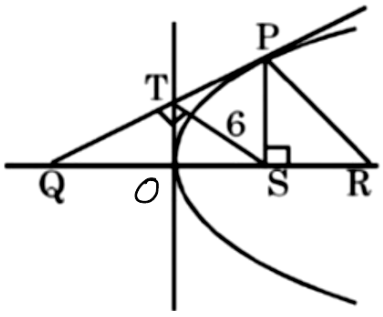
$\Rightarrow\quad\alpha=1,\quad\beta=-2\ \Rightarrow\tan\theta=\alpha+\beta+2=1\Rightarrow\theta=\frac{\pi}{4},\frac{5\pi}{4}$

(ii) $1+\alpha=-2,\quad 1+\beta=1\ \Rightarrow\alpha=-3,\quad\beta=0$

$\Rightarrow\tan\theta=\alpha+\beta+2=-1\ \Rightarrow\theta=\frac{3\pi}{4},\frac{7\pi}{4}$

Total 4 possible values of θ.

6.



As $SP=6=3t^2+3\Rightarrow t=1\ \Rightarrow P=(3,6)$

Tangent at P : 6y = 6 (x + 3) $\Rightarrow\quad x-y+3=0$

$\Rightarrow\quad T=(0,3)\quad\Rightarrow\quad ST=3\sqrt{2}$ also Q = (–3, 0) $\Rightarrow$ QS = 6

As given SR = 6 $\Rightarrow$ R (9, 0)

As we know if tangent and normal at any point P cuts axis at Q & R then SP = SQ = SR

$\Rightarrow$ According. to figure PR is normal

Now Area of PRST = Area of ΔPQR –Area of ΔTQS.

$$= \frac{1}{2} \times PS \times QR - \frac{1}{2} \times OT.QS \ = \frac{1}{2} 6 \times 12 - \frac{1}{2} \times 3 \times 6 = 27 .$$

Also as we know if tangent at P & M meet at point N, then $(SN)^2 = SP.SM$

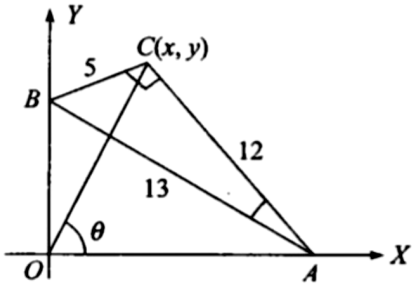
$$25 = 6 \times SM \ \Rightarrow SM = \frac{25}{6} = 3t^2 + 3 \ \Rightarrow t^2 = \frac{7}{18} \Rightarrow t = \pm \frac{\sqrt{7}}{3\sqrt{2}}$$

ordinate of $M = 2at = 2.3.\left(\pm \frac{\sqrt{7}}{3\sqrt{2}} \right)$

$$= \pm \sqrt{14}$$

Sum of squares = 14 + 14 = 28

7.



As $< BCA + < BOA = 90° + 90° = 180° \ \Rightarrow A, O, B, C$ are concyclic

$$\Rightarrow < COB = \frac{\pi}{2} - \theta = < CAB$$

$$\tan\left(\frac{\pi}{2} - \theta \right) = \frac{5}{12} \Rightarrow \cot\theta = \frac{5}{12}$$

Locus of C (x, y) = $\frac{y}{x} = \frac{12}{5}$

Also using ptolemy's theorem :

5. OA + 12. OB = 13.OC

$$\Rightarrow \frac{5}{13} OA + \frac{12}{13} OB = OC \Rightarrow p = \frac{5}{13}, \quad q = \frac{12}{13}$$

8. Divide given differential equation by $xy\left(y^2 - x^2 \right)$.

$$\frac{y\left(y^2 - 2x^2 \right)}{xy\left(y^2 - x^2 \right)} .dx + \frac{x.\left(2y^2 - x^2 \right)}{xy\left(y^2 - x^2 \right)} .dy = 0 \ \Rightarrow \quad \frac{y^2 - x^2 - x^2}{x.\left(y^2 - x^2 \right)} .dx + \frac{y^2 + y^2 - x^2}{y.\left(y^2 - x^2 \right)} .dy = 0$$

$$\Rightarrow \quad \frac{dx}{x} - \frac{x.dx}{y^2 - x^2} + \frac{y\,dy}{y^2 - x^2} + \frac{dy}{y} = 0 \ \Rightarrow \quad \frac{dx}{x} + \frac{dy}{y} + \frac{ydy - xdx}{y^2 - x^2} = 0$$

$$\Rightarrow \quad \ell nx+\ell ny+\frac{1}{2}\ell n\left(y^2-x^2\right)=K \Rightarrow \quad x^2.y^2.\left(y^2-x^2\right)=C$$

$$\Rightarrow \quad f(x)=x^2,\quad g(y)=y^2 \Rightarrow \frac{g(3)}{f(2)}=\frac{g}{4}=2.25$$

9. As plane pass through a common line, their normals will be coplanar.

$$\begin{vmatrix} -1 & \sin\alpha & \sin\beta \\ \sin\alpha & -1 & \sin\theta \\ \sin\beta & \sin\theta & -1 \end{vmatrix}=0$$

$$\Rightarrow \sin^2\theta+\sin^2\alpha+\sin^2\beta+2\sin\alpha.\sin\beta.\sin\theta=1$$

Now $\vec{P}_1\times\vec{P}_2$ is parallel to given line

$$\vec{P}_1\times\vec{P}_2=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & \sin\alpha & \sin\beta \\ \sin\alpha & -1 & \sin\theta \end{vmatrix}=\hat{i}\left(\sin\alpha.\sin\theta+\sin\beta\right)+\hat{j}\left(\sin\theta+\sin\alpha+\sin\beta\right)+\hat{k}$$

$$\left(1-\sin^2\alpha\right)$$

$$\Rightarrow \frac{\sin\alpha.\sin\theta+\sin\beta}{\cos\theta}=\frac{\sin\theta+\sin\alpha.\sin\beta}{\cos\beta}=\frac{1-\sin^2\alpha}{\cos\alpha}=\cos\alpha$$

$$\Rightarrow \sin\theta=\cos(\alpha+\beta),\sin\beta=\cos(\alpha+\theta) \Rightarrow \alpha+\beta+\theta=2n\pi+\frac{\pi}{2}$$

possible values will be $\frac{\pi}{2},\frac{5\pi}{2},\frac{g\pi}{2}$

10. From given matrix,

$$f\left(\frac{1}{x}\right)+f(1)=-x^2f\left((x)+f(1)\right).....(i)$$

$$f\left(\frac{1}{x^2}\right)+f\left(x^2\right)=\left(1-\frac{1}{x^4}\right)f\left(\frac{1}{x^2}\right).....(ii)$$

$$f(x)+f\left(\frac{1}{x}\right)=f\left(\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right).....(iii)$$

$$f(1)+f(x)=-\frac{1}{x^2}f\left(\frac{1+f(1)}{x}\right).....(iv)$$

From (ii), x = 1 $\quad 2f(1)=0\Rightarrow f(1)=0$

$\therefore$ Equation $(i), -x^2f(x)=f\left(\frac{1}{x}\right).....(*)$

Now, $\int_{x}^{1/x} f(t)\,dt = I$ Replace $t \to \frac{1}{t}$

$$\Rightarrow I = \int_{1/x}^{x} f\left(\frac{1}{x}\right)\left(-\frac{1}{t^2}\right)dt = \int_{x}^{1/x} f\left(\frac{1}{t}\right)\frac{1}{t^2}dt$$

$$\Rightarrow 2I = \int_{x}^{1/x}\left(f(t) + \frac{1}{t^2}f(t)\right)dt = 0 \quad \text{from} \quad (*)$$

i.e., I = 0 = f (1).

11. ${}^{7}C_{6}.{}^{8}C_{7}$ (only Rani) $+{}^{8}C_{6}.{}^{8}C_{7}$ (only Radha) $+{}^{8}C_{6}.{}^{8}C_{8}$ (neither Rani nor Radha)

12. Use graph of f (x) = y and y = $\left(\frac{1}{e} - e\right)x + \frac{2}{e}$

13. $x^4 - 7x^2 - 4x + 20 = \left(x^2 - 4\right)^2 + (x-2)^2$

$$x^4 + 9x^2 + 16 = \left(x^2 + 4\right)^2 + x^2$$

Take the curve $y = x^2$. Both square roots can be interpreted as distances.

14. f (2) = 4 _____(1)

$$f\left(2^+\right) = (2+h)^2 + \left(a^2 + ab + 2\right) + (a+b)h$$

$$= a^2 + ab + b \quad \text{_____(2)}$$

$$f\left(2^-\right) = 0 - \left(a^2 + ab + 2\right) + (a+b).(1-h)$$

$$= -a^2 - ab - 2 + a + b \quad \text{_____(3)}$$

from (1) and (2)

$a^2 + ab - a - b + 6 = 0$

$\Rightarrow \quad a + b = 4$ (using – 4)

from (4) a (a + b) + 2 = 0

4a + 2 = 0 $\Rightarrow a = \frac{-1}{2} \quad \Rightarrow b = \frac{g}{2}$

15. $f\left(2^+\right) = f\left(2^-\right) \neq f(2)$

$a^2 + ab + 6 = -a^2 - ab - 2 + a + b \Rightarrow 2a^2 + 2ab - a - b + 8 = 0$ for $a \in R \Rightarrow D \geq 0$

$(2b-1)^2 - 4.2.(8-b) \geq 0 \Rightarrow 4b^2 + 4b - 63 \geq 0$

smallest positive value of b = 4

16. Consider $\int_4^8 \frac{f'(x)}{(f(x))^2}.dx = I$ by substituting f (x) = t, we get

$I = \left(\frac{-1}{f(x)}\right)_4^8 = \frac{-1}{f(8)} + \frac{1}{f(4)} = -2 + 4 = 2$

Now, consider $J = \int_4^8 \left(\frac{f^1(x)}{f^2(x)} + K\right)^2 = \int_4^8 \frac{(f^1(x))^2}{f^4(x)}dx + 2K\int_4^8 \frac{f^1(x)}{f^2(x)}dx + K^2\int_4^8 dx$

$J = 1 + 2K \times 2 + K^2.(8-4) \quad = 1 + 4K + 4K^2$

$\Rightarrow \int_4^8 \left(\frac{f^1(x)}{f^2(x)} + K\right)^2 dx = (2K+1)^2 \quad \Rightarrow \int_4^8 \left(\frac{f^1(x)}{f^2(x)} - \frac{1}{2}\right)^2 dx = 0$

$\Rightarrow$ Integrand , $\left(\frac{f^1(x)}{f^2(x)} - \frac{1}{2}\right)$ must be 0. $\Rightarrow \frac{f^1(x)}{f^2(x)} = \frac{1}{2}$ (always)

Integrating both sides w.r. t x $\frac{-1}{f(x)} = \frac{x}{2} + C$

putting x = 4 $\Rightarrow \frac{-1}{f(4)} = \frac{4}{2} + C \quad \Rightarrow -4 = 2 + C \Rightarrow C = -6$

$\Rightarrow f(x) = \frac{2}{12 = x} \Rightarrow f(6) = \frac{2}{12-6} = \frac{1}{3}$ value of $\frac{1}{5.f(6)} = \frac{3}{5} = 0.6$

17. Given integral can be written as $I = \int_0^1 \left(\prod_{r=4}^{11} 12 - (-x+r)\right).\left(\sum_{r=4}^{11} \frac{2}{12-(-x+r)}\right).dx$

$= \int_0^1 \left(\prod_{r=4}^{11} (12 - r + x)\right).\left(\sum_{r=4}^{11} \frac{2}{12 - r + x}\right).dx$

$= 2\int_0^1 (x+8)(x+7) - - - - (x+1).\left[\frac{1}{x+8} + \frac{1}{x+7} + - - - - - + \frac{1}{x+1}\right].dx$

here integrand is exact derivative of (x + 1) . (x + 2) ---------- (x + 8)

$\Rightarrow I = 2\left[(x+1).(x+2) - - - - (x+8)\right]_0^1 \quad = 2.\left[(2.3.4. - - - - - .9) - (1.2.3. - - - - 8)\right]$

$= 16 \times 8!$

PHYSICS

18. Self induction of a solenoid is $L = \frac{\mu_0 N^2 A}{\ell}$

If x is the length of the wire used to make the solenoid with R as radius of the solenoid, then $x = 2\pi RN$, $A = \pi R^2$

$$L = \mu_0 \left(\frac{x^2}{4\pi^2 R^2} \right) \frac{\pi R^2}{\ell} \Rightarrow x = \sqrt{\frac{4\pi L \ell}{\mu_0}}$$

19. $w = \int PdV$

20. Normal to the mirror is along $\hat{j}$

Along normal to the mirror, $\left(\overline{V_1}\right)_N = \left(2\overline{V_M} - \overline{V_0}\right)_N = 2\left(3\hat{j}\right) - 2\hat{j} = 4\hat{j}\, m/s$

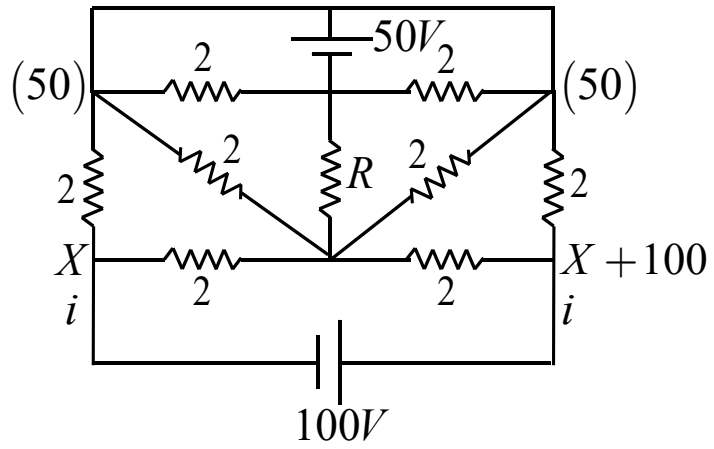
Parallel to the plane of mirror (X–Z plane), $\left(\overline{V_1}\right)_P = \left(\overline{V_0}\right)_P = \left(\hat{i} - 5\hat{k}\right) m/s$

$\therefore \overline{V_1} = \left(\overline{V_1}\right)_N + \left(\overline{V_1}\right)_P = \left(4\hat{j}\right) + \left(\hat{i} - 5\hat{k}\right) = \left(\hat{i} + 4\hat{j} - 5\hat{k}\right) m/s$

21. At y according to Kirchhoff's junction law,

$$\frac{y-x}{r} + \frac{y-x-100}{r} + \frac{y-50}{r} + \frac{y}{r} + \frac{y-50}{r} = 0$$

$5y - 2x = 200$ ……. (i)



Similarly at x, $i = \frac{50-x}{r} = \frac{y-x}{r}$ ……. (ii)

At x + 100, $i = \frac{x+100-50}{r} + \frac{x+100-y}{r}$ ……. (iii)

From Eqs. (ii) and (iii), we get

y – 2x = 50 ……. (iv)

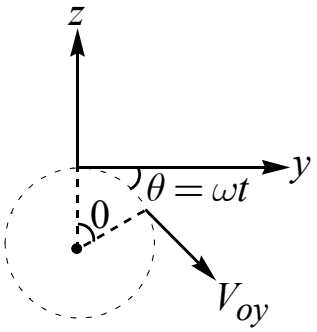
From Eqs. (i) and (iv), y = 37.5V So, current through R is 37.5/18.75 = 2A

22. $\overline{B} = \frac{\mu_0}{4\pi} \frac{q\left(\vec{v} \times \vec{r}\right)}{r^3}$

$$B_0\hat{j} = \frac{\mu_0}{4\pi}\frac{q}{r^3}\left[\vec{v}\times\left(-5\hat{i}\right)\right]$$

This equation becomes true only if $\vec{v}$ has $-\vec{k}$ component or $+\hat{i}$ and $-\hat{k}$ components. Hence options 'B' and 'C' are true.

23. The x–component of velocity, being parallel to magnetic field, shall remain unchanged. The component of velocity perpendicular to x–axis will always have magnitude V_{oy}, and at any time t it shall make an angle $\theta = \omega t$ with y–axis as shown.



$\therefore$ y–component of velocity is $V_{oy}\cos\omega t$ and z–component of velocity along negative z–direction at any time t is $V_{oy}\sin\omega t$. Where $\omega = \frac{qB}{m}$.

24. (a) $i = \frac{dq}{dt}$

$$i = u\int \rho dA = u\int_0^a cr^2 \times 2\pi r dr = 2\pi cu \times \frac{a^4}{4} = \frac{\pi cua^4}{2}$$

(b) $\int EdA = E\times 2\pi r\ell = \frac{\int_0^r c\times r^2 \times 2\pi r\ell dr}{\varepsilon_0}$

$$E = \frac{c\times r^3}{4\varepsilon_0}$$

(c) $B\times 2\pi r = \frac{\mu_0 \times \pi c u r^4}{2}$

$$B = \frac{\mu_0 cur^3}{4}$$

25. Applying energy conservation

$$\frac{1}{2}k\left(\frac{a}{2}\right)^2+\frac{1}{2}2k\left(\frac{a}{2}\right)^2=\frac{1}{2}mv_1^2+\frac{1}{2}Mv_2^2 \qquad \text{..... (i)}$$

By law of conservation of momentum

$-mv_1+Mv_2=0$

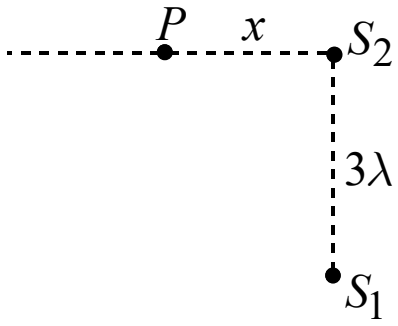
$v_1=4v_2 \qquad \text{..... (ii)}$

$$\frac{1}{2}k\left(\frac{a^2}{4}\right)+\frac{ka^2}{4}=\frac{1}{2}v_1^2+\frac{1}{2}\times 4v_2^2$$

$$v_2^2=\frac{3ka^2}{80}$$

$v_2=3m/s$

26.



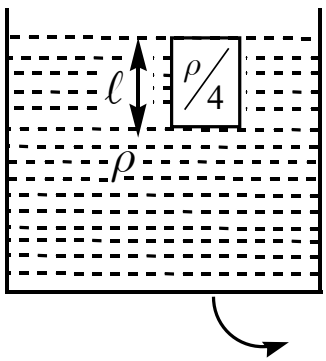
$S_1P-S_2P=\frac{5\lambda}{2}$

$$\sqrt{(3\lambda)^2+x^2}-x=\frac{5\lambda}{2}$$

$$9\lambda^2+x^2=\frac{25\lambda^2}{4}+x^2+5\lambda x \Rightarrow x=\frac{11\lambda}{20}=\frac{11}{20}\times 40\times 10^{-8}\approx 2\times 10^{-7}$$

$K=2$

27. The motion of the cylinder will be simple harmonic. In equilibrium, the block will float with 1/4th of its length submerged. If further pushed by a distance x,



$\left(\frac{\rho}{4}\right)\ell a\times acc^n=-(\rho\ ag)x \quad \Rightarrow W=\sqrt{\frac{4g}{\ell}}$. At equilibrium $V_{\max}=W\times Amplitude$

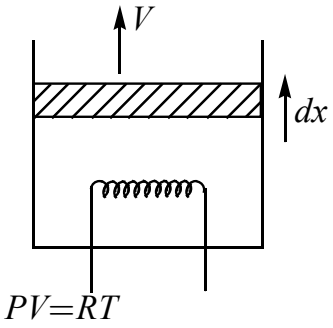
$\Rightarrow V_{\max} = \sqrt{\frac{4g}{\ell}} \times \frac{3\ell}{4} = \frac{3}{2}\sqrt{g\ell} = 3m/s$

28. $dQ = dU + dW$

$dw = pdv = ps\ dx$

$dw = \left(P_0 + \frac{Mg}{s}\right) s\ dx$

$dW = \left(P_0 s + Mg\right) dx$



$dU = \frac{3}{2} RdT$ also $PV = RT \quad \Rightarrow pdv = RdT \Rightarrow psdx$

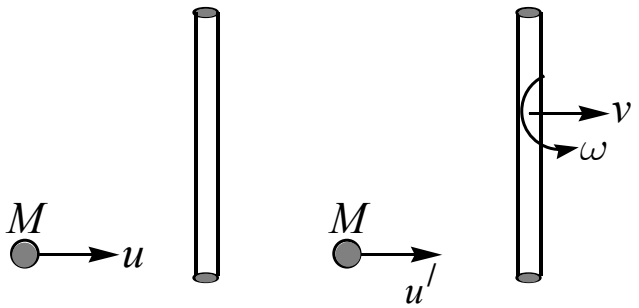
$dU = \frac{3}{2}\left(P_0 + \frac{Mg}{s}\right) s\ dx = \frac{3}{2}\left(P_0 s + Mg\right) dx$

$dQ = \frac{5}{2}\left(P_0 s + Mg\right) dx \Rightarrow v = \frac{2}{5}\frac{dQ/dt}{P_0 S + Mg} \quad \Rightarrow V = 7\ m/s$

29. Power loss = $100 = k\left(50 - 10\right) \Rightarrow k = \frac{10}{4} = \frac{5}{2}$

$\therefore$ At $30^\circ C\ P_{Loss} = k\left(30 - 10\right) = \frac{5}{2} \times 20 = 50W$

30.



By momentum conservation $Mu = Mv + Mu' \qquad u = v + u'$

By angular momentum conservation $M.u\frac{\ell}{2} = Mu'\frac{\ell}{2} + \frac{M\ell^2}{12}.\omega \quad u - u' = \frac{\omega\ell}{6}$

For elastic collision $v + \frac{\omega\ell}{2} - u' = u$ On solving $v = \frac{2u}{5};\ \omega = \frac{12u}{5\ell}$

Now KE of upper half part = $\frac{1}{2}\left(\frac{M}{2}\right)(V_{cm})^2 + \frac{1}{2}.I_{cm}.\omega^2$

$$= \frac{1}{2}\frac{M}{2}.\left[\frac{2u}{5} - \frac{3u}{5}\right]^2 + \frac{1}{2}\frac{\frac{M}{2}.\left(\frac{\ell}{2}\right)^2}{12}\left(\frac{12u}{5\ell}\right)^2 = \frac{Mu^2}{25}$$

31&32

$$P_0 + h_1\rho_1 g + h_2\rho_2 g = \frac{1}{2}\rho_2 v^2 + P_0$$

$$F = \rho a v^2$$

33&34

$$C_{eff} = \frac{15\times 10}{25} = 6\mu F$$

$$q_1 : q_2 = 2:1$$

$$q_2 = \frac{1}{3}q_{total} = 20\ \mu C$$

CHEMISTRY

35. 1) We are given that the sum of the principal quantum numbers is 4: n1 + n2 = 4

2) And their difference is 2: n1 - n2 = 2

Solving these two equations simultaneously:

1) Adding the two equations gives 2n1 = 6, so n1 = 3

2) Substituting n1 = 3 into either equation gives n2 = 1

2) Apply the Rydberg formula for Hydrogen-like species:

The wave number $\left(\tilde{v}\right)$ of the emitted radiation for a hydrogen-like species (like Li^{2+}) is given by: $\tilde{v} = R * Z^2 * \left(1/n2^2 - 1/nl^2\right)$

Where:

1) R is the Rydberg constant.

2) Z is the atomic number (Z = 3 for Lithium).

3) n1 and n2 are the principal quantum numbers of the initial and final levels, respectively.

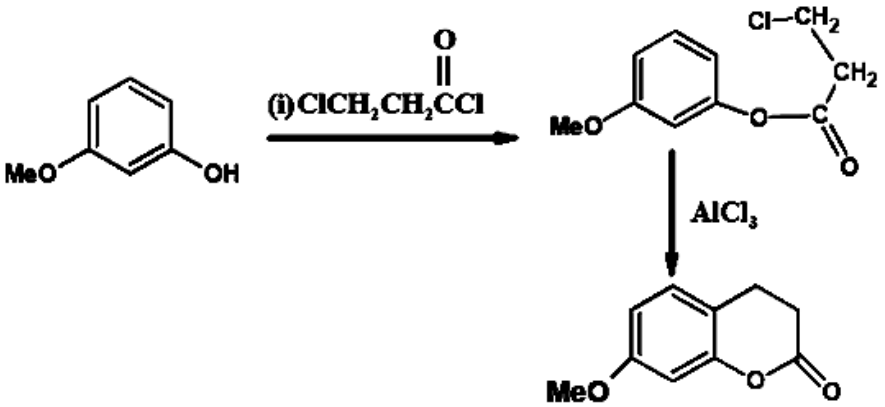
3. Substitute the values and calculate:

$$\tilde{v} = R*(3)^2*\left(1/1^2 - 1/3^2\right)\tilde{v} = 9R*(1-1/9)\tilde{v} = 9R*(8/9)\tilde{v} = 8R$$

36. The sequence 5'-ATGCCTAA-3' has the following base pairs:

1) A-T: 2 hydrogen bonds
2) T-A: 2 hydrogen bonds
3) G-C: 3 hydrogen bonds
4) C-G: 3 hydrogen bonds
5) C-G: 3 hydrogen bonds
6) T-A: 2 hydrogen bonds
7) A-T: 2 hydrogen bonds

Total hydrogen bonds = (2 * 2) + (3 * 3) + (2 * 2) = 4 + 9 + 4 = 17

37. 

38. (a) Incorrect: The reaction is a halogen exchange (metathesis) reaction, not a free radical reaction. Free radical mechanisms involve species with unpaired electrons, which are not involved here.

(b) Incorrect: While *small amounts* of intermediate fluorinated species might form, the reaction's primary and favored outcome is the formation of SF_4. It implies a significant mixture of products, which isn't the case. The reaction is driven towards SF_4.

(c) Incorrect: While SF_4 and AgCl are the *major* products, the stoichiometry *does* matter. You need enough AgF to replace all the Cl atoms in SCl_4. If you have insufficient AgF, you'll likely have some partially fluorinated sulfur chlorides as well. The statement "regardless of stoichiometry" is therefore not entirely accurate.

(d) Correct: This option correctly identifies the driving forces:

1) Stronger S-F bonds: Fluorine's higher electronegativity leads to stronger S-F bonds compared to S-Cl bonds. This provides a thermodynamic driving force.

2) AgCl precipitation: The formation of insoluble AgCl removes products from the reaction mixture, driving the equilibrium towards product formation according to Le Chatelier's principle. This makes the reaction essentially irreversible.

39. A) $\Delta G^o = -nFE^o \Rightarrow -237.39\times10^3 = -2\times96500\times E^o_{cell} \qquad \therefore E^o_{cell} = 1.23\ \text{V}$

C) $\eta = \frac{\left(-\Delta G^o\right)}{\left(-\Delta H^o\right)} = \frac{237.39}{285.8} = 0.8306$ or 83.06%

40. First step is dehalognation and followed by hydrogenation

41. It is likely sulphide salt

42. $t_{av} = \frac{1}{k} = \frac{1}{A.e^{-E_a/RT}} = \frac{1}{\left(1.25\times10^8\ s^{-1}\right)\times e^{-16\times10^3/2\times400}} = 4\,\text{sec}$

43. $\Delta T_f = K_f.m$

For AB_2 solution: $2.55 = 5.1\times\frac{1/\left(x+2y\right)}{20/1000}$ (1)

For AB_4 solution: $1.7 = 5.1\times\frac{1/\left(x+4y\right)}{20/1000}$ (2)

$\therefore$ Atomic mass of A = x = 50

Atomic mass of B = y = 25

44. ----------

45. $3A(g) \rightarrow 2B(g) + 2C(s)$

t = 0	6 atm	0	–
t = 20 min	(6 – x) atm	2/3 x atm	0.05 atm
t = ∞	$\approx$ 0	4 atm	0.05 atm

But from question, P_∞ = 4.05 atm and hence, (4.05 – 4) = 0.05 atm is the vapour pressure of C(s).

Now, $P_{20} = \left(6-x\right)+\frac{2}{3}x+0.05 = 5.05 \Rightarrow x = 2$

$\therefore$ t =20 min = $t_{1/2}$

46. 1. Reaction with Ammonium Thiocyanate:

Iron(III) chloride $\left(FeCl_3\right)$ reacts with ammonium thiocyanate $\left(NH_4SCN\right)$ to form a blood-red complex, which is complex (P). The reaction can be represented (in simplified form, ignoring the exact stoichiometry) as:

$Fe^{3+} + SCN^- \rightarrow \left[Fe\left(SCN\right)_x\right]^{3-x}$ (where x is the number of thiocyanate ligands bound to the iron).

The complex (P) is a high-spin complex. Iron(III) has a d^5 configuration. In a high-spin complex, all five d electrons are unpaired.

2. Spin-only Magnetic Moment of (P):

The spin-only magnetic moment $\left(\mu\right)$ is calculated using the formula:

$\mu = \sqrt{n\left(n+2\right)}$ BM, where 'n' is the number of unpaired electrons.

For complex (P), n = 5 (five unpaired electrons).

$\mu(P) = \sqrt{5}(5+2) = \sqrt{35} \approx 5.92BM.$ So, a ≈ 5.92

3. Reaction with Sodium Fluoride:

Adding excess sodium fluoride (NaF) to complex (P) results in the formation of a colorless complex (Q). Fluoride is a strong field ligand and will replace the thiocyanate ligands. It forms a complex like this:

$$\left[Fe(SCN)_x\right]^{3-x} + F^-(excess) \rightarrow \left[FeF_6\right]^{3-}$$

The complex (Q), $\left[FeF_6\right]^{3-}$, is a high-spin comple

$\mu(Q) = \sqrt{5}(5+2) = \sqrt{35} \approx 5.92BM$. So, b ≈ 5.92

47. Benzaldehyde with formaldehyde gives cross Cannizaro product

48&49

1: Molar Ratio of Styrene to Isoprene

1. Assume 100g of copolymer: This makes the percentage calculations easier. We'd have 60g of styrene and 40g of isoprene.
2. Calculate moles of each monomer:
   1) Moles of styrene = (60 g) / (104 g/mol) = 0.577 moles
   2) Moles of isoprene = (40 g) / (68 g/mol) = 0.588 moles
3. Calculate the molar ratio:
   1) Ratio of styrene to isoprene = (0.577 moles) / (0.588 moles) ≈ 0.98

   Therefore, the chemist should use a molar ratio of approximately 0.98:1 (or very nearly 1:

   1) styrene to isoprene in the initial feed.

   2: Moles of CCl_4 Moles of total monomer: From question 1, in our 100g example, we had approximately 0.577 moles of styrene + 0.588 moles of isoprene ≈ 1.165 moles of total monomer.

1. Number of polymer chains: Since one molecule of CCl_4 terminates one chain, the number of moles of CCl_4 needed will be equal to the number of polymer chains we want. We'll use the relationship between number average molecular weight (Mn), the total mass of polymer, and the number of polymer chains (n): Mn = (Total Mass) / (n)
2. Calculate moles of chains:
   1) n = (Total Mass) / Mn = (10000 g) / (50,000 g/mol) = 0.2 moles of chains.
3. Moles of CCl_4: Since each CCl_4 molecule terminates one chain, we need 0.2 moles of CCl_4.
4. Ratio of CCl_4 to total monomer:

   (Moles of CCl_4) / (Moles of total monomer) = (0.2 moles) / (1.165 moles) ≈ 0.18

Therefore, the chemist should add approximately 0.0017 moles of CCl_4 per mole of total monomer in the initial feed. This very small amount shows how effectively chain transfer agents can control molecular weight.

50&51

$A: NH_3; B: NO; C: NO_2; D: N_2O_3; E: HNO_2; F: N_2; G: CaNCN.$